

- 1 (a) By reference to the direction of propagation of energy, state what is meant by a transverse wave.

.....
 [1]

- (b) A space telescope is designed to detect electromagnetic radiation with wavelengths in the range $12\mu\text{m}$ to $28\mu\text{m}$.

State the region of the electromagnetic spectrum for this radiation.

..... [1]

- (c) A detector on another space telescope detects an electromagnetic wave. The signal from the detector is transmitted to Earth and displayed on an oscilloscope as shown in Fig. 5.1. The frequency of the signal displayed on the oscilloscope is equal to the frequency of the detected electromagnetic wave.

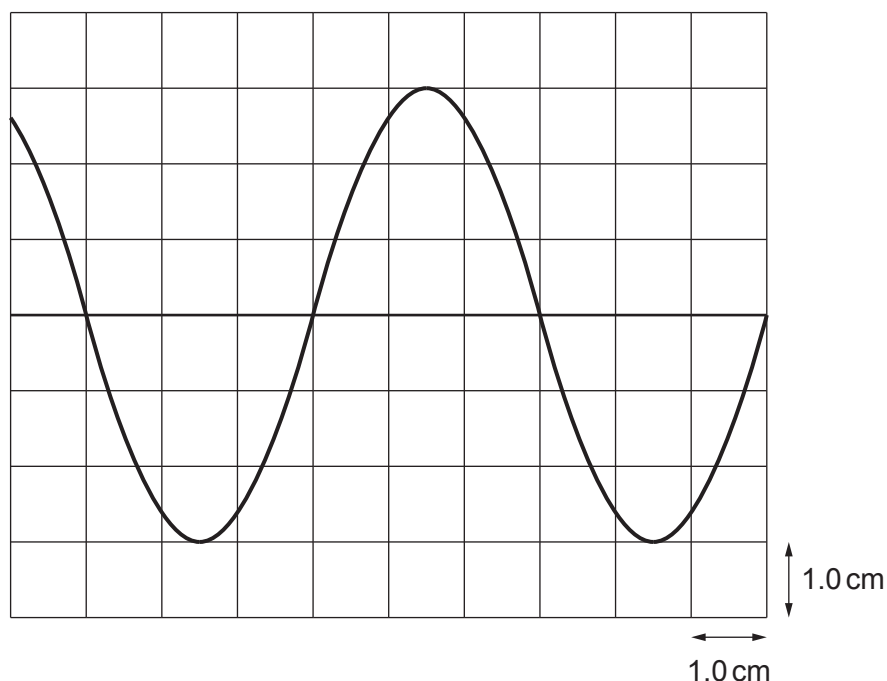


Fig. 5.1

The time-base setting on the oscilloscope is $5.0 \times 10^{-15} \text{ s cm}^{-1}$.

Calculate the wavelength of the detected electromagnetic wave.

wavelength = m [3]

[Total: 5]

- 2 A progressive transverse wave travelling from left to right is shown at an instant in time in Fig. 4.1.

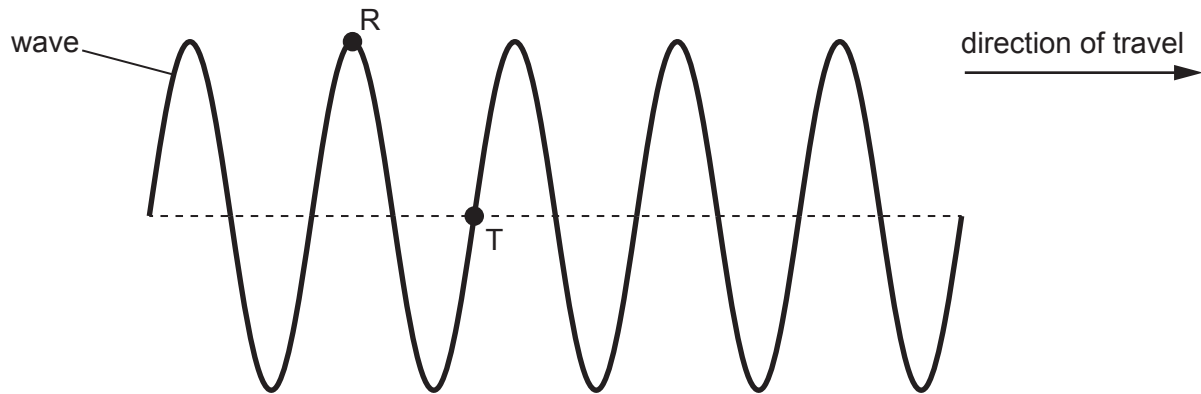


Fig. 4.1

R and T are points on the wave.

- (a) State the phase difference between the points R and T.

phase difference = ° [1]

- (b) On Fig. 4.1, draw an arrow at point T to show the direction of movement of point T at the instant shown. [1]

- (c) The horizontal distance between R and T is 0.62 cm, as shown in Fig. 4.2.

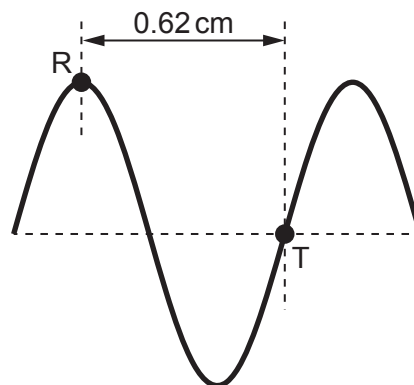


Fig. 4.2 (not to scale)

The speed of the wave is 0.27 m s^{-1} .

Calculate the frequency of the wave.

frequency = Hz [3]

- (d) A source of sound waves of frequency 780 Hz is on a rotating platform. The speed of the source is 39 m s^{-1} .

The sound is detected by an observer that is a large distance from the rotating platform, as shown in Fig. 5.2.

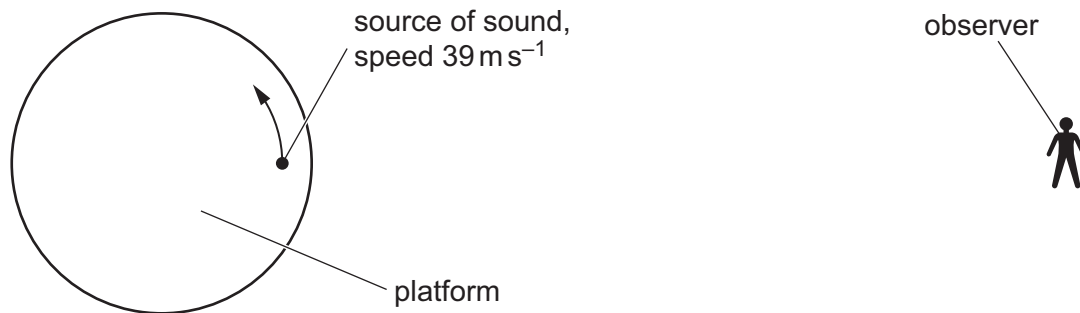


Fig. 5.2 (not to scale)

- (i) The speed of sound in air is 320 m s^{-1} .

Calculate the maximum frequency of the sound detected by the observer.

maximum frequency = Hz [2]

- (ii) At time $t = 0$, the observer detects the sound emitted by the source when it was in the position shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with t of the frequency f of the sound detected by the observer for one complete rotation of the platform. Calculations are not required.



Fig. 5.3

[2]

[Total: 8]

- 3 (a) Coherent visible light of a single frequency is incident normally on a double slit. This produces a pattern of bright and dark interference fringes on a screen, as illustrated in Fig. 6.1.

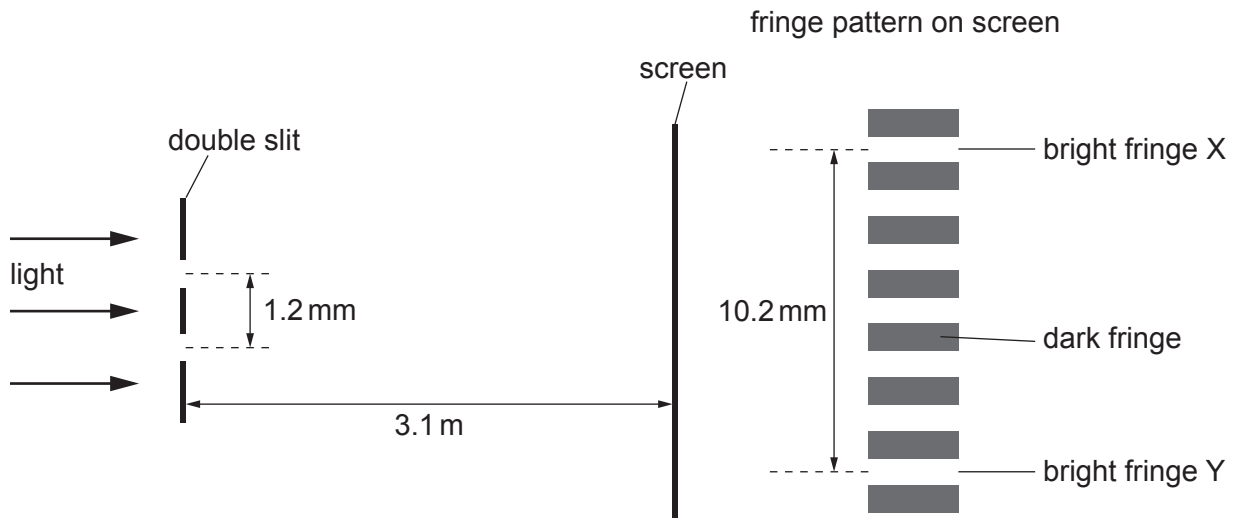


Fig. 6.1 (not to scale)

There are seven bright fringes.

- (i) Explain how the pattern of bright and dark interference fringes is formed.

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..... [3]

- (ii) The distance between the centres of bright fringe X and bright fringe Y in the pattern is 10.2 mm. The slit spacing is 1.2 mm. The distance from the slits to the screen is 3.1 m.

Calculate the wavelength of the light incident on the slits.

wavelength = m [3]

- (iii) The light is replaced by different visible light with a shorter wavelength.

State how the new fringe separation will compare to the original fringe separation.

..... [1]

- (b) The wave is a water wave produced by a dipper S_1 attached to a vibrator in a ripple tank.

An identical dipper S_2 is attached to the same vibrator. The two dippers produce an interference pattern on the water in the tank, as shown in Fig. 4.3.

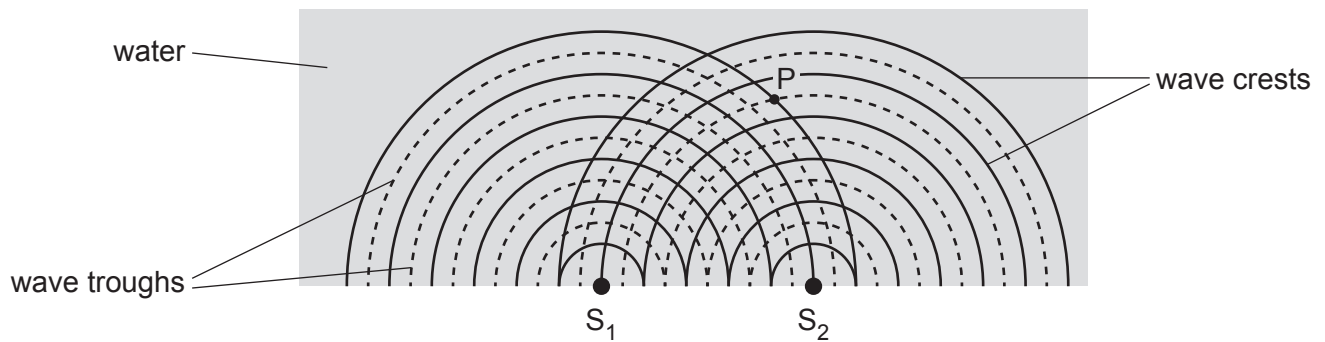


Fig. 4.3 (not to scale)

The wave crests from each source are represented by solid lines on Fig. 4.3 and the wave troughs are represented by dashed lines.

At point P in Fig. 4.3, the wave from S_1 has the same amplitude A as the wave from S_2 .

Describe and explain the amplitude of the resultant wave at point P.

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..... [3]

[Total: 8]

- 4 Two coherent sources X and Y of microwaves of frequency 2.5×10^{10} Hz are a distance of 0.18 m apart in a vacuum, as shown in Fig. 6.1.

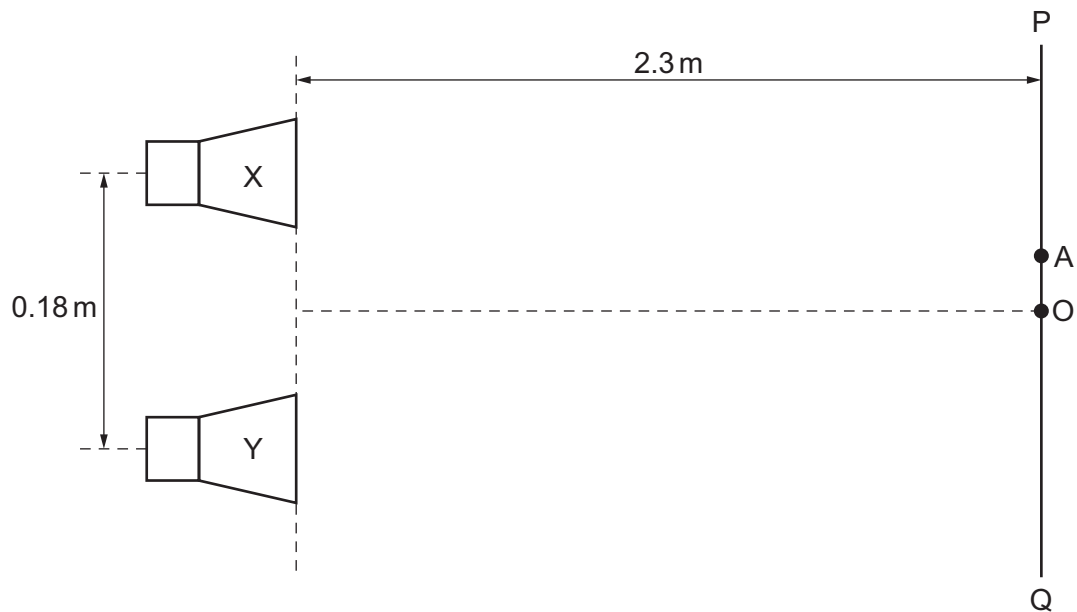


Fig. 6.1 (not to scale)

There is a phase difference of 90° between the waves emitted at the two sources.

A microwave detector moves along the line PQ, which is parallel to the line joining the two sources and 2.3 m away from it.

Point O is on the line PQ at a position that is equidistant from the two sources.

Point A is the position on line PQ where the intensity of the microwaves is the greatest.

- (a) (i) Explain why the position of greatest intensity is **not** at point O.

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 [2]

- (ii) On Fig. 6.1, draw a cross (x) to show the position of the point on line PQ where the intensity minimum that is the closest to point O occurs. Label this point B. [2]

- (b) (i) Show that the wavelength of the microwaves is 0.012 m.

[2]

- (ii) For point A on line PQ, determine the difference in the distances Δx travelled by the microwaves from X and the microwaves from Y.

$\Delta x = \dots\dots\dots$ m [1]

- (iii) Use the formula for the double-slit interference of light to calculate the distance between adjacent intensity maxima on line PQ.

distance = $\dots\dots\dots$ m [2]

[Total: 9]

- 5 Light of a single frequency is incident normally on a diffraction grating. An interference pattern of bright and dark fringes forms on the semicircular screen shown in Fig. 6.1.

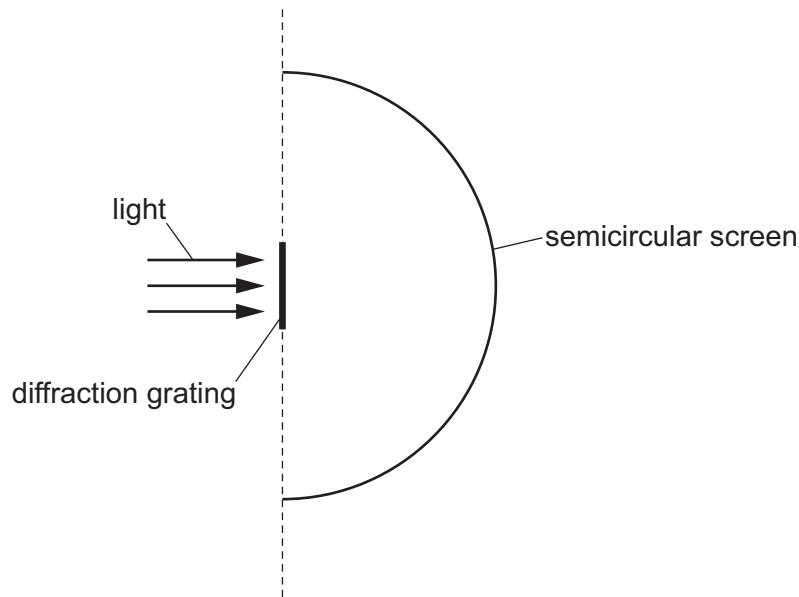


Fig. 6.1 (not to scale)

The light has wavelength 520 nm.

The separation of the lines in the grating is 3.8×10^{-6} m.

- (a) Determine the total number of bright fringes formed on the screen.

number of bright fringes = [3]

- (b) The light is replaced with red light of a single frequency.

- (i) State whether the frequency of the red light is greater than, less than or the same as the frequency of the original light.

..... [1]

- (ii) State and explain the effect of this change on the number of bright fringes formed on the screen. A calculation is not required.

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 [2]

[Total: 6]

- 6 (a) A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

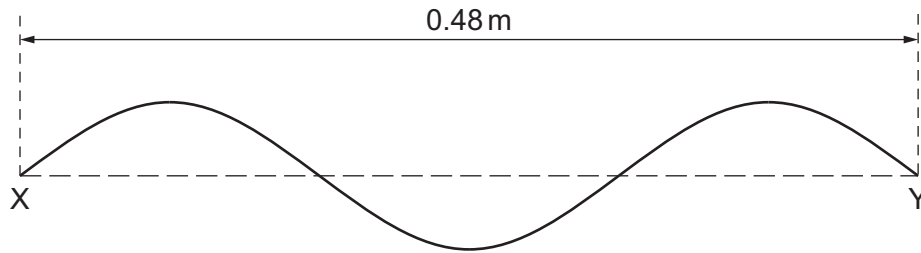


Fig. 5.1

The speed of the wave on the string is 1400 ms^{-1} .

- (i) On Fig. 5.1, draw a cross (×) at **one** position that is a node and another cross at **one** position that is an antinode. Label the node N and the antinode A. [1]
- (ii) Show that the wavelength of the wave produced is 0.32 m. Explain your reasoning. [1]
- (iii) Calculate the frequency of the wave. [1]

frequency = Hz [2]

- (b) A stationary wave is formed on a stretched string AB, as shown in Fig. 6.2.

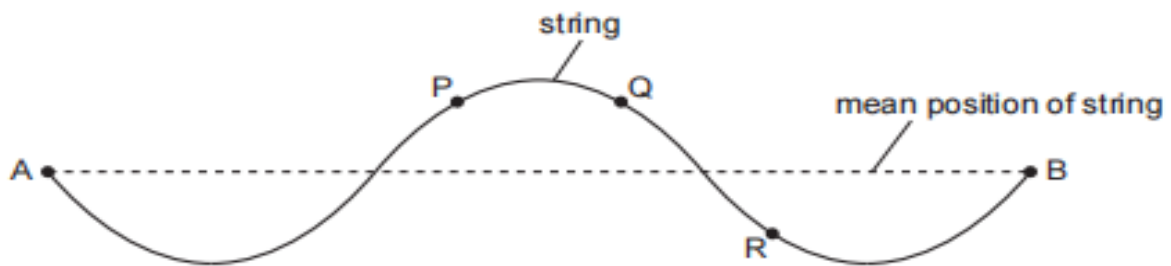


Fig. 6.2

P, Q and R are points on the string.

- (i) On Fig. 6.2, draw a cross (×) to show the position of a node. [1]
- (ii) State the phase difference between P and Q. [1]
- phase difference = ° [1]
- (iii) State the phase difference between P and R. [1]
- phase difference = ° [1]

[Total: 10]

- 7 A stretched string PQ has length 1.2 m. One end of the string is attached to a vibration generator and the other end is attached to a wall, as shown in Fig. 5.1.

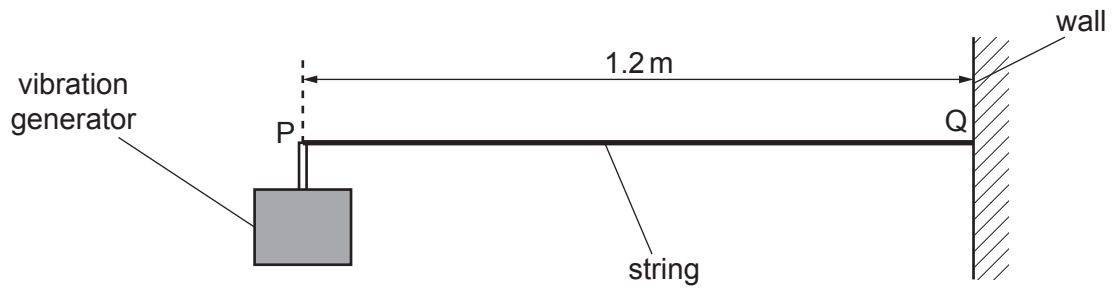


Fig. 5.1

The vibration generator is switched on and a stationary wave is formed on the string. The string is shown at one instant of time in Fig. 5.2.

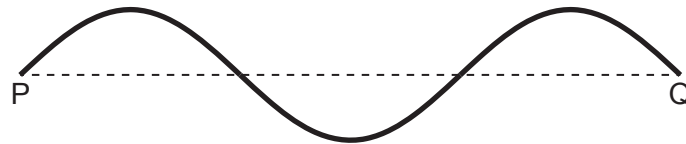


Fig. 5.2

- (a) Explain how a stationary wave is formed between the vibration generator and the wall.

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..... [2]

- (b) Calculate the wavelength of the stationary wave shown in Fig. 5.2.

wavelength = m [1]

- (c) Fig. 5.3 shows the stationary wave at time $t = 0$ when all points on the wave are at their maximum displacements.

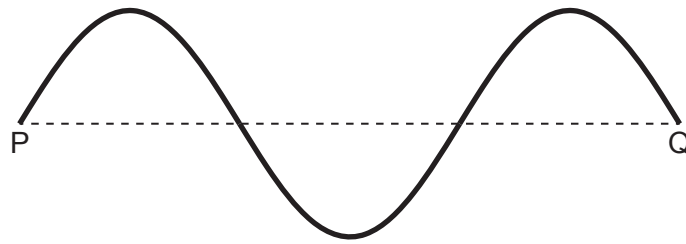


Fig. 5.3

The period of the wave is 0.16 s.

On Fig. 5.3, sketch the shape of the stationary wave at time $t = 0.24$ s.

[2]

- (d) Points R and T on the string are a horizontal distance of 0.30 m apart and in the positions shown in Fig. 5.4.

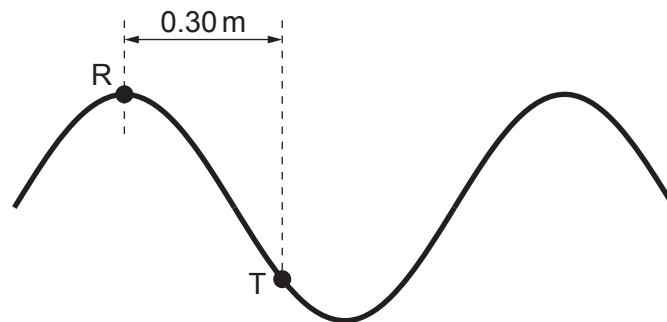


Fig. 5.4

State the phase difference between the oscillations of points R and T.

phase difference = ° [1]

- (e) Calculate the speed of the progressive waves on the stretched string.

speed = ms^{-1} [2]

[Total: 8]

- 8 (a) With reference to the direction of transfer of energy, compare the oscillations of transverse and longitudinal progressive waves.

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..... [2]

- (b) A pipe is open at one end and closed at the other with a piston. The piston can slide freely and is at a distance of $4.5 \times 10^{-2} \text{ m}$ from the open end of the pipe.

A loudspeaker is positioned near the open end of the pipe and emits a sound wave of a single constant frequency. A stationary wave is formed in the pipe, as illustrated in Fig. 4.1.

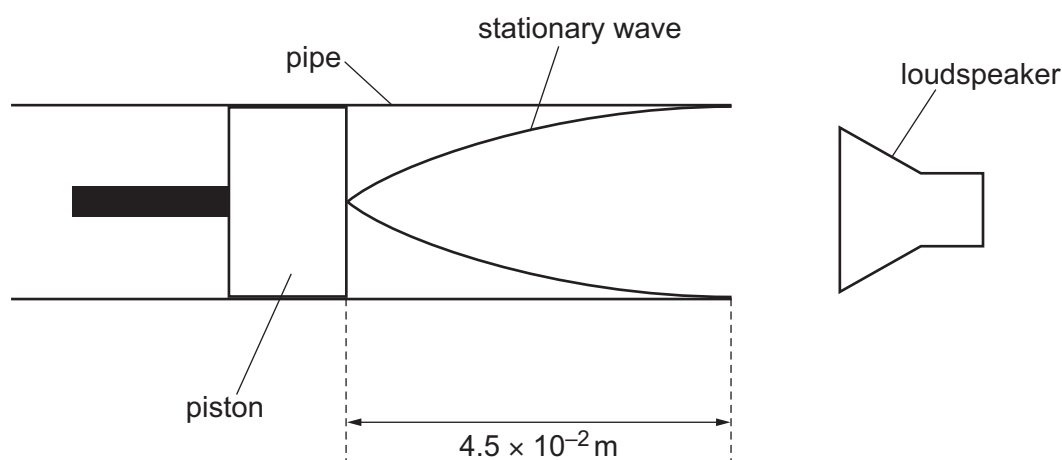


Fig. 4.1

- (i) On Fig. 4.1, draw a letter A at the position of an antinode. [1]

- (ii) The speed of sound in air is 340 ms^{-1} .

Determine the frequency of the sound wave.

frequency = Hz [3]

- (iii) The piston is moved to the left. The frequency of the sound wave emitted by the loudspeaker is then changed so that a stationary wave is formed with same number of antinodes as in Fig. 4.1.

State and explain the change that is made to the frequency of the sound wave.

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..... [2]

[Total: 8]